

FIFO DESIGN Using Verilog

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Presented here is a first-in, first-out (FIFO) design using Verilog that is simulated using ModelSim software. In this

article, we design and analyse FIFO using different read and write logics. We have considered 64 inputs, each having 32-bit data.

FIFO design

FIFO is an approach for handling program work requests from queues

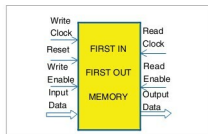


Fig. 1: FIFO memory I/O

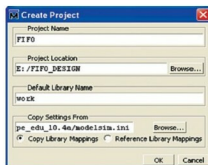


Fig. 2: Create Project window



Fig. 3: Add items to the Project window

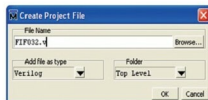


Fig. 4: Create Project File window

SINGLE PORT FIFO PIN DESCRIPTION		
Pin name	I/O	Function
Input Data<31:0>	Input	To enter the data when Write Enable (active high) and Write Clock are asserted.
Output Data<31:0>	Output	To receive the data when Read Enable(active high) and Read Clock are asserted.
Write Clock	Input	To perform write operation from input port.
Read Clock	Input	To perform read operation from output port.
Write Enable	Input	To perform write (active high) operation from input port.
Read Enable	Input	To perform read (active high) operation from output port.
Reset	Input	For reset the FIFO

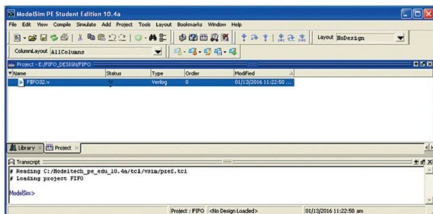


Fig. 5: Workspace window

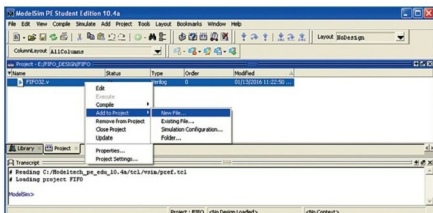


Fig. 6: Adding new files